

A Hybrid Deep Learning Framework for Human Activity Recognition Using Image and Video Datasets

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Abstract

HAR is a fundamental research area in computer vision, finding applications in applications like surveillance systems, intelligent environments, emotion analysis, and behavior understanding. This work proposes a hybrid deep learning framework for HAR on two different but related datasets: one image-based, which is used to train a CNN for spatial feature learning, and another video-based, containing five classes of human activities for the temporal classification of activities. Our approach first uses a CNN to extract discriminative spatial features from image frames. These are then used in two kinds of classification strategies: using a CNN-based classifier alone and a hybrid CNN-Decision Tree (CNN-DT) model. The experimental results demonstrate that the hybrid CNN-DT model provides improved performance over the only CNN model on the video dataset by achieving 80.71% accuracy against 78.57%. It was illustrated that the combination of deep feature extraction with classical machine learning classifiers enhances the recognition accuracy and robustness for video-based human activity recognition.

Keywords: Human Activity Recognition, Convolutional Neural Network, Decision Tree, Hybrid Model, Video Classification, Deep Learning

1. Introduction

HAR is now a fast-growing research domain within computer vision and artificial intelligence due to the wide range of real-world applications it could be used for: intelligent surveillance systems, healthcare monitoring, human–computer interaction, smart homes, and assistive technologies. The ultimate goal of HAR is to automatically detect and recognize human actions and activities from images and videos. Precise human activity recognition plays an important role in enhancing the intelligence of the system and establishing natural interaction between humans and machines.

The early methods for human activity recognition were heavily based on handcrafted features and traditional machine learning techniques. Feature extraction was typically performed using Histogram of Oriented Gradients, Scale-Invariant Feature Transform, and optical flow, among other features, before classification through algorithms such as Support Vector Machines and Decision Trees. While these representations worked reasonably well in laboratory settings, their general performance was poor under real-world conditions because illumination, background clutter, occlusion, viewpoint variations, and complicated motion patterns can easily degrade the extracted features [1]. These limitations eventually called for more robust and automated feature extraction methods.

With the advent of deep learning techniques, the Convolutional Neural Networks (CNNs) have found applications in the realm of visual recognition. The CNNs have the capability to learn hierarchical and discriminative function mappings from the raw input data in the image, thus doing away with the necessity of hand-crafted feature design. This, in turn, has improved the performance of CNN models in applications including image classification tasks, object detection, facial analysis, and action recognition [2]. Specifically, the CNN models have found application in facial emotion recognition tasks, in which the spatial facial characteristics are critical in recognizing the diverse emotional states. The previous research has revealed the superiority of deep CNN architectures over the conventional approaches in the process of emotion recognition [3].

Since image-based recognition involves mainly spatial information, a new temporal aspect is added when performing video-based human activity recognition.

This increases the level of complexity. Spatial information in videos, as well as information from a temporal aspect, plays a vital role in recognizing dynamic human actions like hand gestures. The recognition of hand gestures has attracted considerable attention because of its relevance in human computer interaction, virtual reality, and assistive technology. Researchers have been able to use CNN-based methods effectively for recognizing hand gestures based on spatial information from videos, which has been quite promising [4].

Nevertheless, pure CNN models may not necessarily be effective in capturing the temporal dependency existing in video data for certain datasets, particularly those that contain a restricted number of samples. In an effort to overcome these limitations, combined models that integrate deep models with classifiers and sequence models have also been introduced. These combined models seek to take advantage of the feature extraction property in CNN models and make use of Decision Trees and Support Vector Machines (SVMs) for effective decision-making as well as utilize LSTM networks for temporal modeling [5]. These models have also exhibited better stability and performance in human activity recognition.

The key contributions of this research are as follows:

- 1) This work presents a unified framework for image-based emotion recognition and video-based hand gesture recognition.
- 2) The implementation and comparison of various deep and hybrid models for human activity recognition using video.
- 3) Extensive experimental and graphical analysis to illustrate the benefits of hybrid models compared to deep learning methods alone.

2. Related Work

Human Activity Recognition (HAR) has gained increasing attention in the past few years because it has many usages, including surveillance, healthcare, human-computer interaction, and intelligent systems. In past methods for HAR, features manually designed, including Histograms of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), and optical flow features, were primarily used, and traditional classifiers, including Support Vector Machines (SVMs) and Decision Trees, were applied to classify human activities from the extracted features. Although past methods were providing satisfactory results, they had many limitations because they were very feature-engineering dependent and had less generality on real-world problems.

With the increasing advancements in deep learning, Convolutional Neural Networks (CNNs) have emerged as prominent solutions in image-based recognition

problems. CNNs possess the ability to learn hierarchical features from raw images. Various researches prove the efficacy of CNNs on facial emotion recognition tasks by correctly classifying various classes of emotions, including anger, happiness, sadness, fear, and surprise [6], [7]. These research studies prove the impressive performance of deep CNNs over traditional machine learning approaches for image-based emotion classification problems.

Temporal information is very important in video-based activity and gesture recognition. To utilize spatial and temporal features simultaneously, there have been attempts to incorporate spatiotemporal deep learning approaches such as 3D Convolutional Neural Networks (3D-CNNs). 3D-CNNs process multiple video frames simultaneously, and there have been very prominent results in detecting dynamic gestures and actions among them [8]. These approaches tend to be computationally expensive and are subjected to the risk of overfitting in smaller databases.

To counter such drawbacks, hybrid approaches consisting of CNNs and recurrent neural networks such as Long Short-Term Memory Networks (LSTM) have emerged. The CNN-LSTM network architecture combines CNNs for capturing spatial information and LSTMs for capturing time-based relationships among video frames. A number of research initiatives have observed better accuracy and convergence with CNN-LSTM hybrid networks for recognizing human actions from videos [9], [10]. Moreover, a combination of machine learning classification algorithms like SVM/Decision Trees with CNN-based feature extraction has been observed to carefully classification accuracy while avoiding misclassifications for multi-class classification tasks [11].

Recent research using surveys highlights that a hybrid approach using deep learning with both spatial and temporal components outperforms individual models significantly [12]. Notwithstanding these recent advancements, most research that currently exists is restricted to either images or video recognition. There is very little research on a holistic approach that tests simultaneous recognition by different hybrid models on both images and videos.

In light of the discussed observations, the aim of this study is to explore the CNN-based learning process in the recognition of image-based emotional expressions, as well as the application of several deep and hybrid approaches, such as 3D-CNN, CNN-Decision Tree, CNN-LSTM-SVM, in the gesture recognition process based on videos.

3. Proposed Methodology

This paper presents a hybrid framework based on deep learning models for Human Activity Recognition

(HAR) tasks based on images as well as videos. The proposed framework includes data processing, extracting features using deep learning models, hybrid classification, and evaluation. The main aim is to demonstrate the efficacy of deep models, namely Convolutional Neural Networks, and hybrid models in extracting features.

3.1 Image Dataset Processing and CNN Architecture

In image-based recognition, a facial emotions dataset comprising seven classes of emotions, including angry, disgust, fear, happy, neutral, sad, and surprise, is employed. Nonetheless, all the images are normalized to have the same dimensions to be processed by the convolutional neural network model after being subjected to data augmentation methods like rotation, flip, and scaling. The detailed class-wise distribution of the image dataset is presented in Table 3.1.

Class	Images
Angry	5,920
Disgust	5,920
Fear	5,920
Happy	11,398
Neutral	8,166
Sad	6,535
Surprise	5,920
Total	49,779

Table 3.1: Class-wise distribution of the image dataset

A CNN neural network is used to learn the discriminative spatial information from the facial images. A CNN model is composed of several layers of convolutions followed by max pooling layers to reduce the spatial dimension without losing important information. Fully connected layers are used to learn the higher-level representation of the images followed by the Softmax function to provide multi-class outputs. CNN is useful in image-based classification because of its ability to model the hierarchical structure of images [2], [6].

The performance of the trained CNN model is then measured by using some standard evaluation metrics, which include accuracy, per class accuracy, confusion matrix, precision, recall, as well as F1-score. The graphical evaluation of test accuracy and class accuracy provides information about the training dynamics of the model.

3.2 Video Dataset Description and Preprocessing

In video-based activity recognition, a dataset of gestures is used, which contains 140 videos distributed across five classes of gestures: fist, four, me, one, and small. Each video will be decomposed into its individual

frames, and a fixed number of frames is selected to maintain temporal consistency among samples. Frames are resized and normalized before passing them through the deep learning models.

The video data introduces temporal dependencies that are not effectively modeled with only image-based CNNs. So, several deep and hybrid architectures are used to model both spatial and temporal information.

3.3 3D-CNN Model for Video Recognition

The initial video-based model that is employed here is the 3D Convolutional Neural Network (3D-CNN). The difference in this model is that it involves convolution operations in the spatial as well as the temporal dimension. This is helpful in learning the dynamics of the gesture from the frames. However, this model consumes higher computational power and larger amounts of data [13].

3.4 Hybrid CNN–Decision Tree Model

To make it more robust for classification, a CNN-DT model is employed. The methodology is designed such that CNN features are utilized for spatial information extraction in video frames, and a Decision Tree classifier is employed for classification. Decision Trees are known for their interpretability and efficiency in dealing with non-linear decision boundaries. Overfitting and instability are avoided by using a combination of CNN and Decision Trees, and boosting is gained [11].

3.5 Hybrid CNN–LSTM–SVM Model

The third and most complex architecture is a CNN-LSTM-SVM hybrid system. In this architecture, CNN is utilized for extracting spatial features from the video frames; these features are further processed using LSTM to capture temporal dependencies between several frames of videos. Features obtained are finally categorized using a Support Vector Machine (SVM) classifier; this classifier is famous for its robust generalization properties even in higher-dimensional feature spaces.

This is a hybrid approach combining the strengths of CNN in spatial tasks, LSTM in dealing with the time domain, and SVM in classification. This hybrid model has proved to be highly efficient in handling HAR in video datasets [8], [9].

4. Experimental Setup

For the experimental analysis of the framework proposed, the programming language employed is Python. This language is widely used for machine

learning as well as deep learning due to its flexibility-related advantages along with the availability of an extensive library. For the implementation of deep learning concepts through the help of the TensorFlow and Keras libraries, the construction of the CNNs takes place effectively. Other machine learning methods along with the evaluation metrics related to them can be found through the scikit-learn library. [14] ,[15]

The image dataset is used for training as well as testing the CNN model for emotion recognition in the faces. The images in all cases undergo pre-processing before the training is conducted in the neural network. The CNN model is optimized with back-propagation algorithms after the categorical cross-entropy loss function has been applied to the model for training, following the training principles for deep learning algorithms [2].

For video-based gesture classification tasks, the dataset is split into training and testing sets for a fair assessment of generalization performance. Frames from videos are processed prior to being fed into various deep learning as well as hybrid approaches. Three different network architectures are used for evaluation: a 3D Convolutional Neural Network (3D-CNN), a hybrid model involving a CNN and a Decision Tree classifier, and a hybrid model involving a CNN, an LSTM network, and an SVM classifier.

The performance of all the models implemented is assessed using the standard classification measures: accuracy, precision, recall, and F1 score. In this work, accuracy is the primary measure to evaluate the overall classifying performance of the models. Other measures give insights into the class performance and robustness. This method of evaluating the model is standard in human activity recognition research to guarantee accurate results. [8] , [13].

5. Results and Discussion

5.1 Image-Based Dataset Results

The CNN model was initially trained on image-based dataset to authenticate its ability to represent spatial features. The CNN model has been able to acquire spatial features that are discriminatory, which are extracted from images of different classes of activities. The training and validation accuracy plots were able to demonstrate that there are no serious overfitting issues, ensuring effective generalization. These spatial features will now be reused for classifying videos.

5.2 Video-Based Dataset Result

First, for the video-based dataset having five human activity classes, three different models were tested to examine their performance:

1) 3D Convolutional Neural

The 3D-CNN model was trained end-to-end on the video clips to incorporate the spatial and temporal information present in the clips. This model acts as a deep learning baseline for recognizing spatiotemporal activities.

2) CNN + Decision Tree (Hybrid Model

In this method, a CNN was employed to extract spatial features from the video frames, and then the features were fed to the Decision Tree classifier for the final classification of the activity. In this hybrid model, the target is to optimize the boundary and the classification error regarding visually similar activities.

3) CNN + SVM + LSTM (Hybrid Model)

The third model adopts the CNN-based spatial feature extraction technique and the Support Vector Machine classification method, and it also incorporates the Long Short-Term Memory networks to address the temporal dependence issue. The combination of these techniques could help tap the benefits of the CNN-based visual features, the Support Vector Machine classification, and the Long Short-Term Memory networks.

5.3 Model Comparison

Accuracy comparison of different models trained and tested on the video dataset can be summarized in the following table:

Model	Dataset type	Accuracy (%)
3D-CNN	Video	78.57
CNN + Decision Tree (Hybrid)	Video	80.71
CNN + LSTM +SVM	Video	80.71

Table 2: Accuracy comparison of proposed models on the video dataset.

The comparative analysis in Table 2 shows in that the hybrid model consistently outperforms the standalone CNN.

5.4 Graphical Analysis

The graphs for the accuracy and losses for both image and video datasets are provided for the training and validation.

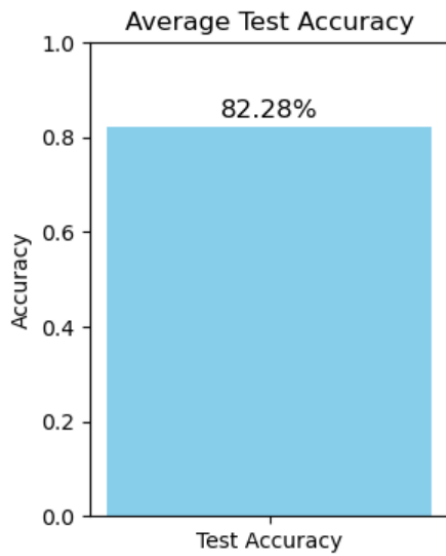


Figure 5.1: Overall test accuracy of CNN on image dataset

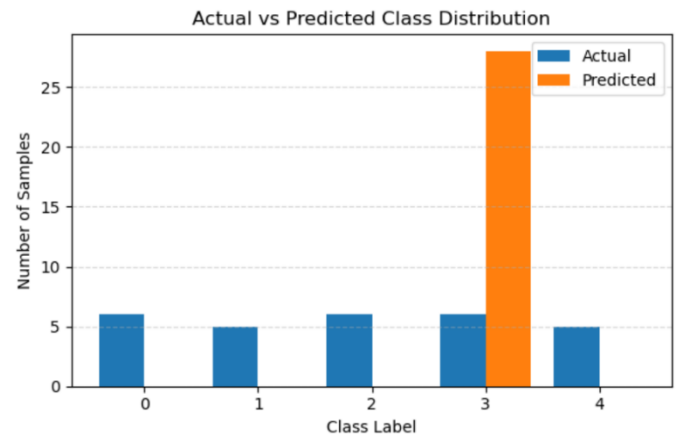


Figure 5.4: Actual vs predicted class distribution for video dataset

Findings are further validated by the graphical outputs and the comparative analysis of all video-based models, as illustrated in Figure 5.5.

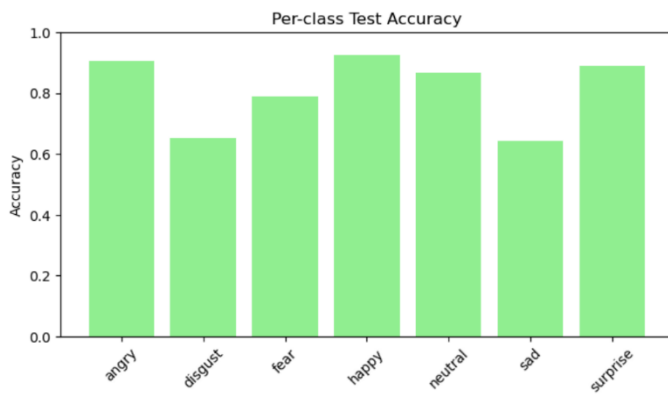


Figure 5.2: Per-class accuracy of CNN on image dataset

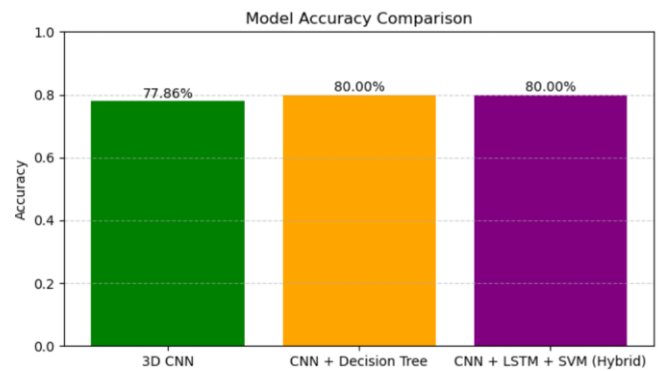


Figure 5.5: Accuracy comparison of different models on the video dataset

The graphs for the image dataset represent uniform learning and convergence for the CNN approach. Likewise, the graphs for the video dataset emphasize enhanced stability and accuracy through the combined CNN- LSTM- SVM method as shown in Figure 5.3. The actual vs predicted class distribution is shown in Figure 5.4.

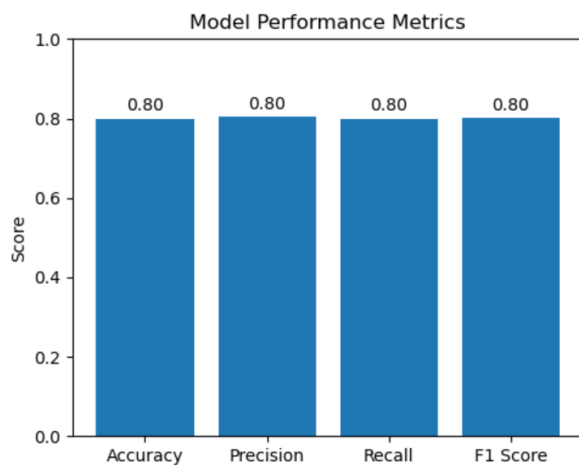


Figure 5.3: Performance metrics of the hybrid CNN–LSTM–SVM model on video dataset

5.4 Graphical User Interface (GUI) Results

In order to ensure the real-world usability of the proposed human activity recognition method, it has been coupled with a graphical user interface (GUI). With the help of the GUI tool, users are able to input their sample image or video example in order to display the corresponding class labels provided by the trained CNN-based models.

In addition, the GUI interface displayed in Figure 5.6 illustrates the image-based emotion recognition system result to identify the emotional classes of the faces recognized through the input images using the CNN model. Comparable to the previous statement, Figure 5.7 displays the GUI result validating the video-based gesture recognition system to identify the recognized activities' classes from the input video sequences.

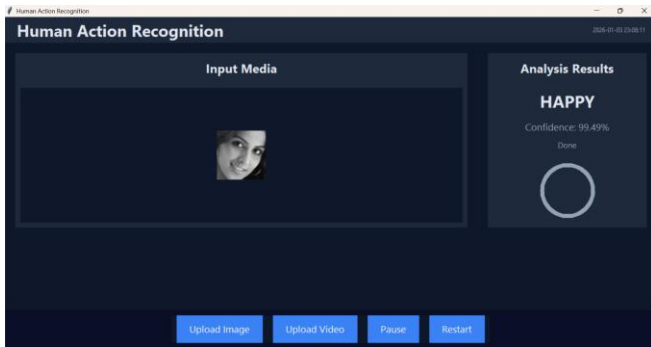


Figure 5.6: Graphical User Interface for image-based emotion recognition

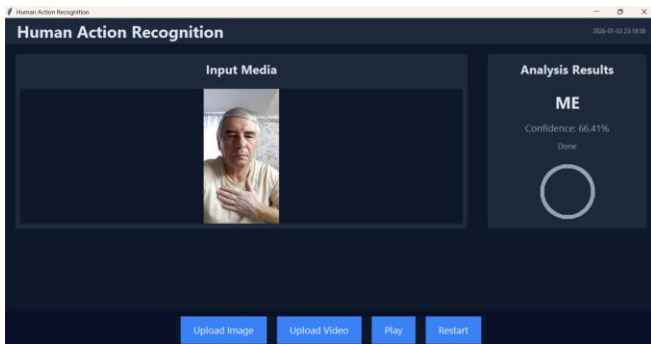


Figure 5.7: Graphical User Interface for video-based gesture recognition

6. Conclusion

The hybrid model CNN and Decision Trees was introduced in the study as a framework for human activity recognition from images and videos. The results showed that the hybrid model outperforms the only CNN model. Future studies may be performed using LSTM or 3D-CNN models.

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